

Title: Data Preparation and Feature Engineering for Course Offering Recommendation System (CORS)

1. Introduction

This section outlines the methodology used to transform raw historical course offering data into a labeled dataset suitable for training a machine learning model. The objective is to predict the likelihood of a specific course being offered in a specific term ($y=1$) or not ($y=0$).

2. Data Sources

- **Historical Grades/Offerings:** A CSV export containing course sections, enrollment numbers, passing rates, and grades from Fall 2020 to Spring 2025.
- **Course Catalog (To be added):** A list of all valid active courses (used to identify electives vs. cores and Math vs. CS).
- **Prerequisite Graph (To be added):** A directed graph or adjacency matrix representing course dependencies.
- **Study Plan (To be added):** A structured representation (JSON) of the ideal semester-by-semester path for students.

3. Data Cleaning and Preprocessing

- **Flattening:** The raw source file was a hierarchical report (containing merged cells and subtotals). We applied forward-filling to propagate Term and Course values to section rows and removed summary/total rows to obtain a flat transaction list.
- **Feature Selection (Raw):** Grade distribution columns (A through WP) were dropped as they are granular outcomes of the course, whereas the prediction relies on the *demand* and *capacity*. We retained Total Enrolled, Passing Grades, and Failed Grades.
- **Aggregation:** The raw data is at the *Section* level (e.g., CSC201 had 5 sections). The Machine Learning model predicts at the *Course* level. Data was aggregated by summing enrolled/passed/failed students per Course per Term.

4. Dataset Construction (The "master_ml_table")

The final dataset is constructed using a Cartesian Product approach to handle **Negative Sampling**:

- *Positive Samples ($y=1$):* Rows representing courses that actually appeared in the historical schedule.
- *Negative Samples ($y=0$):* We generated rows for courses that exist in the catalog but were *not* offered in a specific term. This is crucial for the model

to learn patterns of "non-offering" (e.g., a Fall-only course not appearing in Spring).

5. Feature Engineering

The following features are computed for every row (Course C, Term T, Year Y):

- **Identifies:** Year, Term, Course_Code (Used for mapping, not training).
- **Temporal/Demand Features:**
 - `passed_prereqs_last_2_semesters`: The sum of students who passed the immediate parents of Course C in the 2 previous terms. *Rationale: High volume in prerequisites indicates high demand for the successor.*
 - `failed_students_prev`: The number of students who failed Course C in the previous offering. *Rationale: Failures create a backlog of "repeaters".*
 - `fail_ratio`: The historical difficulty of the course.
- **Curriculum Features:**
 - `study_plan_alignment`: Binary (1/0). Returns 1 if the current Term T matches the suggested term in the standard study plan.
 - `is_math`: Binary (1 if Math department, 0 otherwise).
 - `is_elective`: Binary (1 if the course is a defined elective, 0 if core).

6. Target Variable

- `is_offered`: Binary. 1 if the course had >0 enrollment in that term, 0 otherwise.
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